

3 Carbonic anhydrase is a globular protein found in red blood cells.

(a) (i) Explain why this protein is described as globular.

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(ii) State the function of carbonic anhydrase in red blood cells.

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- (b) A protein such as carbonic anhydrase is coded for by a gene. A gene forms part of a DNA molecule.

Fig. 3.1 is a diagram of a small section of a DNA molecule.

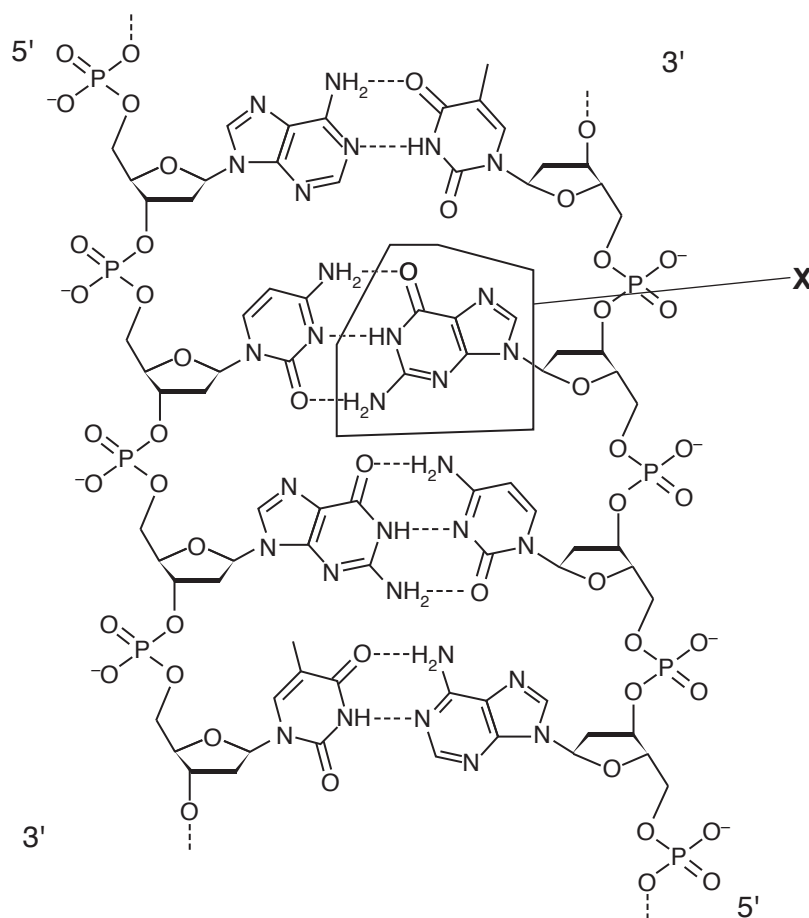


Fig. 3.1

- (i) Identify the base **X** and state the evidence in Fig. 3.1 that supports this identification.

base **X**

evidence from Fig. 3.1

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- (ii) The section of the DNA molecule in Fig. 3.1 is part of a gene coding for a polypeptide.

Base **X**, shown in Fig. 3.1, is located in an exon on the strand of DNA that is transcribed during protein synthesis. A mutation that results in the deletion of base **X** will affect the polypeptide produced.

Explain how this deletion may affect the polypeptide produced during protein synthesis.

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- (iii) Gene mutations can occur in either introns or exons.

Suggest the effect of a gene mutation in an intron.

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[Total: 12]